Review changelog: batch p9-f (Phase 9 lessons 11-14)

Reviewer date: 2026-10-22. Sandbox: react / react-dom 19.3.1, vitest 3.2.7, jsdom, RTL, zod 4.6.7. Scratch code: `/Users/todorbra/Projects/ReactLearning/tools/sandbox/scratch/p9-f/` (`ssr.test.tsx`, `form.test.tsx`, `form2.test.tsx`, `replay*.test.tsx`, `enter.test.tsx`, `a11y.test.tsx`, `z.test.ts`, `rt.test.ts`, `b4.tsx` / `inert.tsx` typechecks, `so2/` server-only alias). Saved docs: `scratch/p9-f/src/` (react.dev) and the session scratchpad (Next, Zod, TanStack, W3C, Google, web.dev).

All ⚠/verify/unverified markers in the four lessons are removed (0 remaining); each was either confirmed, corrected, or replaced by an explicit hedge.

Totals (approx.): confirmed 62, corrected 24, hedged/unresolved 9.

Lesson 11: Progressive enhancement

Claim	How checked	Result	Change
`renderToString(<form action={fn}>)` output	Executed (`ssr.test.tsx`)	**Corrected/clarified**: emits `action="javascript:throw new Error('React form unexpectedly submitted.')"` plus a one-per-response inline `<script>`; `method` is dropped (dev warning "Cannot specify a encType or method for a form that specifies a function as the action")	§5.4 example no longer claims `method="post"` gives a fallback; added explanation + note
String action	Executed	Confirmed: `action="/x" method="post"` emitted verbatim	Noted in §5.4
Server Function reference action	Executed with a simulated `$$FORM_ACTION` reference	Confirmed: `action`, `method="POST"`, `encType="multipart/form-data"`, hidden inputs; `formState` option exists in server+client bundles; `<!--F!-->` / `<!--F-->` markers in source	§3.2 expanded (marked as implementation detail)
Client-function form pre-hydration ("✘ no handler yet")	Source read + executed `replay.test.tsx`	**Corrected**: the inline script captures the submit (preventDefault, FormData incl. submitter, `document.$$reactFormReplay`) and React replays it on hydration; with JS disabled nothing happens. Verified end to end in jsdom (action received the data after `hydrateRoot`)	Table row, §3.4 rewritten, B4 note
Event replay of discrete events	Source (`discreteReplayableEvents` list in react-dom 19.3.1)	List confirmed from source. **Unresolved in execution**: my jsdom harness did not reproduce click replay on a dehydrated Suspense boundary (`replay2.test.tsx`: handler attached after hydration but the queued click was not re-fired)	§3.4 states the list as "read from source", with an explicit "could not reproduce" note
`permalink` semantics	react.dev useActionState + server-functions pages	Confirmed; added the two documented conditions (same component/permalink on destination; no effect once interactive)	§2.4, A5
Reset of uncontrolled fields after action	Executed (`form.test.tsx` A1, A2, A4)	Confirmed, including when action *returns* an error object; controlled input keeps value	§5.2 note, linked to lesson 13
`useFormStatus` must be in a child	Executed (C) + react.dev	Confirmed (same-component `pending` always false)	none
Zod `.email()`	zod.dev v4 + executed	**Corrected**: `z.string().email()` deprecated; `z.email()` does not trim/lowercase first, so example became `z.string().trim().toLowerCase().pipe(z.email().max(254))` (verified output)	§5.2 code + note
`<details name>`, popover, dialog support	MDN pages exist (HTTP 200); compat tables not machine-read	**Hedged**	Wording rewritten; added Invoker Commands mention
Playwright `javaScriptEnabled`	playwright.dev/docs/emulation	Confirmed	removed "unverified"
MDN links (validation, form, 303)	HTTP 200	Confirmed	removed "unverified"
Skew protection	Not checked	**Hedged** ("some hosts/frameworks offer it")	D2

Lesson 12: Auth boundaries

Claim	How checked	Result	Change
CVE-2025-29927 middleware bypass	GitHub advisory GHSA-f82v-jwr5-mffw	Confirmed + completed: `x-middleware-subrequest`, CVSS 9.1, fixed 15.2.3 / 14.2.25 / 13.5.9 / 12.3.5, 11.x workaround, Vercel-hosted auto-protected	§2.4
Middleware "constrained runtime, no DB"	Next 16.4 proxy docs	**Corrected**: Next 16 renames middleware to `proxy`, defaults to Node.js runtime; docs still recommend optimistic checks only	§2.4
Layout checks don't re-run	Next auth guide ("Due to Partial Rendering...")	Confirmed	cited
Server Actions CSRF (Origin vs Host, allowedOrigins)	Next data-security guide + serverActions config page	Confirmed + extended: POST-only, `X-Forwarded-Host`, wildcards, missing `Origin` is allowed with a warning, runs in production, config under `experimental.serverActions`	§2.6, §5.3, §10, D3
React `cache()` scope	react.dev/reference/react/cache	Confirmed (per server request, Server Components only)	§3.1
Closure variable encryption	Next data-security guide	Confirmed (auto-encrypted, key per build, `NEXT_SERVER_ACTIONS_ENCRYPTION_KEY`, "don't rely on encryption alone")	§3.3
Taint APIs status	react.dev pages + Next guide	Confirmed still experimental; `experimental.taint`; "only protects against simple mistakes"	§3.3
`cookies()` async	Next cookies reference	Confirmed (async since 15.0.0-RC; sync access removed in 16 per upgrade guide/codemod text)	§5.1 comment
IDOR/BOLA is OWASP API #1	OWASP API1:2023 page; OWASP Top 10:2025 redirect	Confirmed for the 2023 edition; no newer edition found. **Hedged** ("current edition I could find")	§3.2
Cookie `__Host-` rules	MDN cookies	Confirmed	none
Auth.js / Better Auth / Clerk	authjs.dev, better-auth.com, Next auth guide	Confirmed: authjs.dev banner "The Auth.js project is now part of Better Auth"; Better Auth blog (Jul 2026) "Better Auth is joining Vercel"; Next guide lists Auth0, Better Auth, Clerk, NextAuth.js, etc. **Unresolved**: migration guidance / support terms not checked	New library-landscape paragraph in §7.2 (+ JWT-vs-session-claims note)
React Server Components RCE and follow-ups	react.dev Dec 3 and Dec 11 2025 posts; CHANGELOG	Confirmed: CVE-2025-55182 (CVSS 10, 19.0/19.1.0/19.1.1/19.2.0 -> 19.0.1/19.1.2/19.2.1); DoS CVE-2025-55184/67779, source exposure CVE-2025-55183; 19.0.3/19.1.4/19.2.3 incomplete, fixed 19.0.4/19.1.5/19.2.4; hardening through 19.2.7. **Unresolved**: CVE-2026-23864 not found on react.dev	§8; hedged in text
`server-only` in Vitest	Executed (`so2/`)	Confirmed: default import throws; `resolve.alias` to `empty.js` makes it a no-op	C2 note
C3 `safeReturnTo`	Executed (`rt.test.ts`)	**Bug found and fixed**: `/.//evil.com` returned `//evil.com` (open redirect) because `new URL()` collapses `..`; added `startsWith("//")` guard (all test inputs now return `/` or a safe path)	C3 code + explanation
Zod `z.string().uuid()`	zod 4	**Corrected** to `z.uuid()` (3 places)	code
Paywall structured data	Google paywalled-content doc	Confirmed (`isAccessibleForFree: false`, `hasPart` + `cssSelector`)	§9
Framework cache defaults	Next 16 upgrade/caching docs	**Hedged** (changed in 15 and 16)	§3.4
RLS, NEXT_PUBLIC_, deep links	Next docs, HTTP 200 checks	Confirmed	markers removed

Lesson 13: Forms, actions and mutations

Claim	How checked	Result	Change
Reset after success, incl. returned error object	Executed (A1-A4)	Confirmed; throwing action goes to error boundary (confirmed); `onReset` fires on automatic reset (confirmed, matches 19.3 notes)	§2.5

Claim	How checked	Result	Change
Async-transition limitation after `await`	react.dev useTransition (19.3) + executed (E)	Confirmed: renders `v=0 pending=true`, `v=1 pending=true`, `v=1 pending=false`	§2.3
`useActionState` queueing	react.dev + executed (B)	Confirmed: 3 submits -> 3 sequential calls with prev 1,2,3, 2 renders; thrown error cancels queue (docs)	§3.2
Render sequence for single submit	Executed	`state=0 pending=true` then `state=1 pending=false`	§3.2
`useOptimistic` rollback / no flicker	Executed (D, D3, D4)	**Corrected**: no flicker only if the real update is in the same transition; plain `setState` after `await` gave an intermediate duplicate frame (`a,x(tmp)` / `a,x,x(tmp)` / `a,x`); with `startTransition` it is `a,x(tmp)` / `a,x`; failure case reverts	§3.3, §4
Optimistic warning text	Executed	**Corrected**: exact text on 19.3.1 is lowercase "outside a transition or action..."	table, §10
`requestFormReset`	Exported (source), executed in transition, react.dev page 404	Confirmed available; **no reference page** (hedged)	table row
Enter double submit	Executed with user-event + HTML implicit-submission rule	**Corrected**: disabled default button blocks Enter (1 invocation for 3 presses); no-button form ran 3 times	§4
Submitter name/value in FormData; multi-intent	Executed (G)	**Corrected**: button with function `formAction` loses `name` (React overrides, warns) and handler gets no `intent`; plain submitter contributes `intent`	§5.7 code: removed `name/value` from `formAction` button and `encType` (React sets it; warns)
Next `revalidatePath/Tag`, `updateTag`, `refresh`	Next 16.4 docs	**Corrected/extended**: `revalidateTag(tag, profile)` required second arg (one-arg deprecated), `updateTag` (Server Actions only), `refresh()`	§3.6
Next body size limit	serverActions config page	Confirmed: default 1MB, `serverActions.bodySizeLimit`	§5.7
Zod 4 error API	Executed + zod.dev	`z.flattenError`, `z.treeifyError`, `error.issues` confirmed; `.flatten()` / `.format()` deprecated	§5.2 code + note
TanStack Query retries/offline/variables-based optimistic	TanStack docs	**Corrected**: mutations don't retry by default (opt-in); paused offline mutations resume in order; "via the UI" variant confirmed	table, §5.8
B4 TypeScript catches `(formData) =>`	tsc (`b4.tsx`)	Confirmed (TS2769)	none
`redirect` inside try/catch	Next redirect doc consulted; lesson text unchanged	Confirmed	none
File uploads / FormData `File`	Not executed	Hedged by principle (client-declared `type`)	none
React Hook Form / TanStack Form status	Not checked	**Unresolved** (lesson does not make claims about them)	none
IETF Idempotency-Key draft status	Not fetched	**Unresolved**	hedged text

Lesson 14: Accessibility and SEO

Claim	How checked	Result	Change
WCAG 2.2 criteria	W3C WCAG 2.2 text	Confirmed: 2.4.11 Focus Not Obscured (Minimum) AA, 2.5.7 Dragging Movements AA, 2.5.8 Target Size (Minimum) AA, 3.2.6 Consistent Help A, 3.3.7 Redundant Entry A, 3.3.8 Accessible Authentication (Minimum) AA, 4.1.1 Parsing "Obsolete and removed"	Table row rewritten with numbers/levels; "valid parseable" wording corrected

Claim	How checked	Result	Change
Metadata hoisting	react.dev title/meta/link + `renderToString`	Confirmed; **Corrected**: multiple `<title>` are all emitted; React says behavior is undefined (no "last/closest wins")	§2.7
`useId` format	react.dev 19.2 post + executed	**Corrected**: `:r:` (19.0), `«r»` (19.1), `_r_` (19.2+); server output looked like `_R_0_`	line notes
`inert` typing/support	Executed + tsc	Confirmed	marker removed
Next route announcer	Next accessibility docs	Confirmed (title, then h1, then pathname)	§3.4
Next streaming metadata & bots	generateMetadata docs	Confirmed (HTML-limited bots block; `htmlLimitedBots`)	§2.6
Soft 404 with streaming	Next not-found docs	Confirmed and extended: 404 before streaming; 200 after, with injected `noindex`	§5.5
Google JS SEO / dynamic rendering	Google docs	Confirmed: crawl/render/index, evergreen Chromium, dynamic rendering page marked Deprecated	§2.5
JSON-LD `<` escape	Next JSON-LD guide	Confirmed	§5.5
CWV thresholds, INP replaced FID	web.dev vitals/INP	Confirmed (2.5 s / 200 ms / 0.1; INP successor of FID). The exact "March 2024" date was not found on the fetched pages (not stated in lesson)	§3.5, §9
Next metadata API (`viewport` export, sitemap/robots/opengraph-image)	generateMetadata docs	Confirmed	bullet in §5.5
StrictMode focus-effect misfire	Executed	Confirmed (focus fired on initial mount)	note
Automated tools find "a third"	Not re-fetched	**Hedged** (estimates vary)	§2.8
EAA from mid-2025	Not fetched	**Hedged** (28 June 2025 from memory; verify scope)	§9
jest-axe / vitest-axe / jsx-a11y / @axe-core/playwright status	npm registry	Confirmed versions; vitest-axe 0.1.0 (Jan 2025), jsx-a11y 6.10.2 (Oct 2024, ESLint <=9 peer)	tooling workflow
Legacy React a11y doc; react.dev/learn/accessibility	curl	legacy 200, react.dev 404	resource table
WAI-ARIA 1.2/1.3, APG, "Using ARIA"	HTTP 200 only	Not content-checked; lesson makes no version-specific claim	**Unresolved** (minor)

Most significant corrections

1. Lesson 11: client-function form actions are NOT dead pre-hydration: React's inline script queues and replays submits after hydration; and `method="post"` on a function-action form is dropped, so §5.4's "fallback" was wrong.
2. Lesson 12: `safeReturnTo` allowed `/..//evil.com` (open redirect); Next 16 `proxy` runs on Node; CVE-2025-29927 and RSC CVE details filled in.
3. Lesson 13: `useOptimistic` no-flicker only holds when the real update is in the same transition; multi-intent `name` / `value` with function `formAction` does not work; Next 16 `revalidateTag` signature and `updateTag`.